



ENGINEERING MATHEMATICS-II

MATLAB

Department of Science and Humanities

- MATLAB has the **rand** and **randn** functions for generating uniformly and normally distributed random variables, respectively.
- **X=rand** returns a random scalar drawn from the uniform distribution on the open interval (0,1)
- **X=rand(n)** returns an n -by- n matrix of uniformly distributed random numbers on the open interval (0,1)
- **X=randn** returns a random scalar drawn from the standard normal distribution
- **X=randn(n)** returns an n -by- n matrix of normally distributed random numbers

Examples

```
>> X=rand(1,4)
```

```
X = 0.8147    0.9058    0.1270    0.9134
```

```
>> X=rand(1,4)
```

```
X = 0.6324    0.0975    0.2785    0.5469
```

```
>> X=rand(2,2)
```

```
X =    0.1576    0.9572
```

```
    0.9706    0.4854
```

Examples

```
>> X=rand(3)
```

```
X = 0.8003 0.9157 0.6557
```

```
    0.1419 0.7922 0.0357
```

```
    0.4218 0.9595 0.8491
```

Examples

```
>> X=randn(1,2)
```

```
X = 1.0933 1.1093
```

```
>> X=randn(3)
```

```
X = 0.0774 -0.0068 0.3714
```

```
-1.2141 1.5326 -0.2256
```

```
-1.1135 -0.7697 1.1174
```

Bernoulli Distribution:

- The Bernoulli distribution is a discrete probability distribution with only two possible values for the random variable.
- Each instance of an event with a Bernoulli distribution is called a Bernoulli trial.
- Clearly, X is a discrete random variable, with probability mass function $p(x)$

defined by $p(0) = P(X = 0) = 1 - p$

$$p(1) = P(X = 1) = p$$

$p(x) = 0$ for any value of x other than 0 and 1

Bernoulli Distribution PDF

- $\text{binopdf}(x, n, p)$ computes the binomial probability density function at each of the values in x using the corresponding number of trials n and probability of success for each trial in p .

Bernoulli Distribution PDF



Use binopdf to compute the pdf of the Bernoulli distribution with a probability success of 0.90.

```
>> p = 0.90;
```

```
>> x = 0:1;
```

```
>> y=binopdf(0:1,1,p)
```

```
y = 0.1000  0.9000
```


Bernoulli Distribution PDF

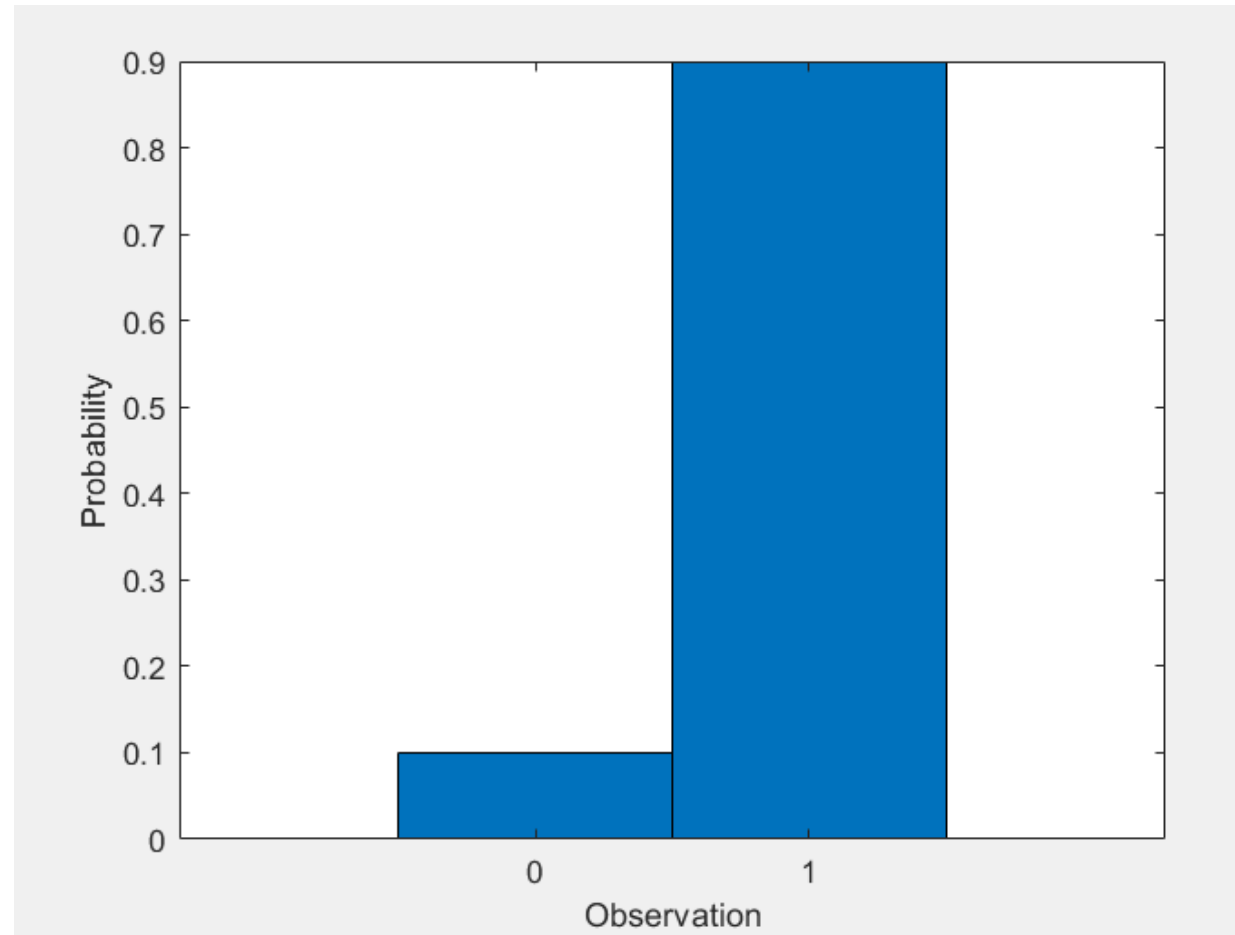
Now, plot the PDF with bars of width 1.

```
>> figure
```

```
bar(x,y,1)
```

```
xlabel('Observation')
```

```
ylabel('Probability')
```



Bernoulli Distribution CDF

```
>> p = 0.90;
```

```
>> y = binocdf(-1:2,1,p)
```

```
y = 0    0.1000    1.0000    1.0000
```

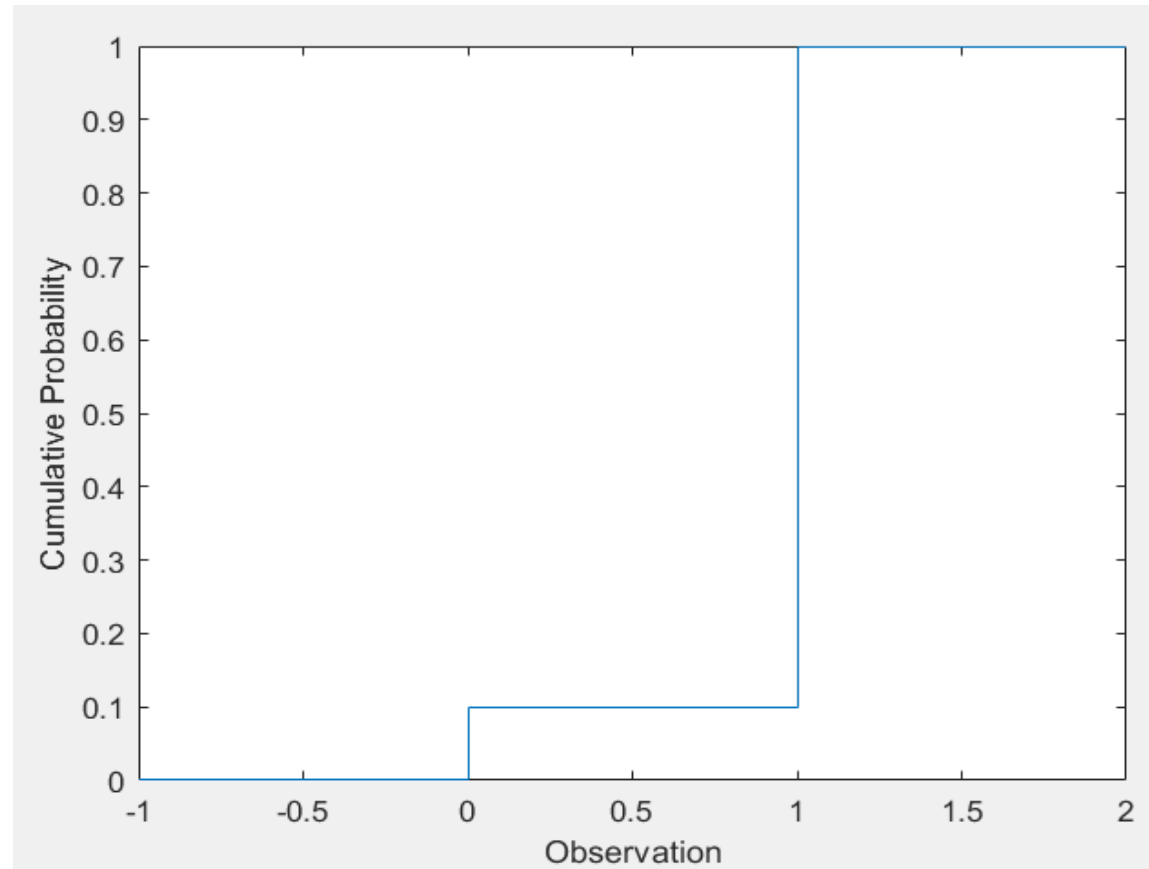
Plotting of CDF:

```
>> figure
```

```
stairs(-1:2,y)
```

```
xlabel('Observation')
```

```
ylabel('Cumulative Probability')
```



Binomial Distribution

- The binomial distribution is a two-parameter family of curves.
- The binomial distribution is used to model the total number of successes in a fixed number of independent trials that have the same probability of success, such as modeling the probability of a given number of heads in ten flips of a fair coin.

Binomial Distribution PDF

Compute the PDF of the binomial distribution with 15 trials and the probability of success 0.5.

```
>> x = 0:15
```

```
>> y = binopdf(x,15,0.5)
```

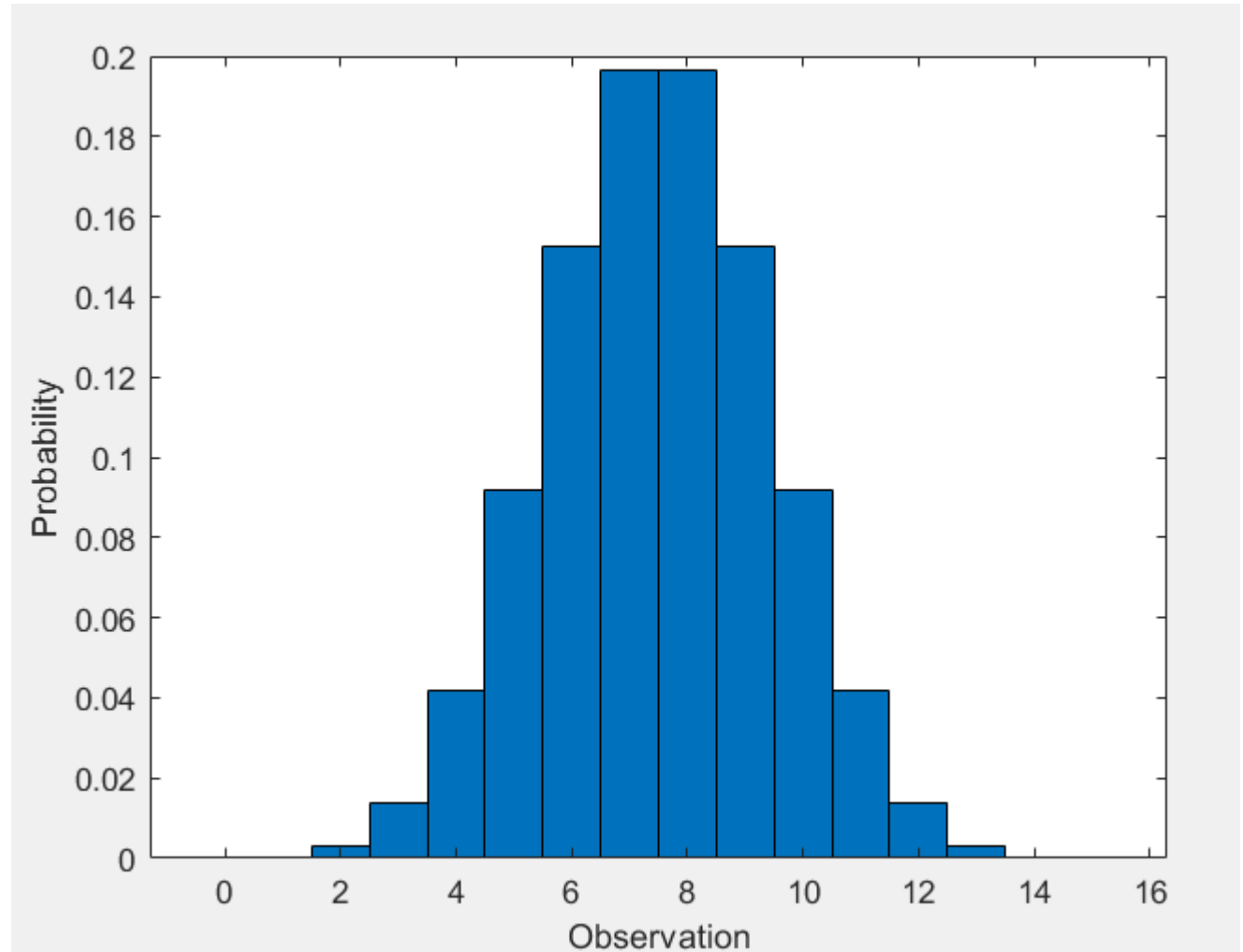
Plotting of PDF with bars of width 1:

```
>> figure
```

```
bar(x,y,1)
```

```
xlabel('Observation')
```

```
ylabel('Probability')
```



Binomial Distribution CDF

Compute the CDF of the binomial distribution with 15 trials and the probability of success 0.5.

```
>> x = 0:15
```

```
>> y = binopdf(x,15,0.5)
```

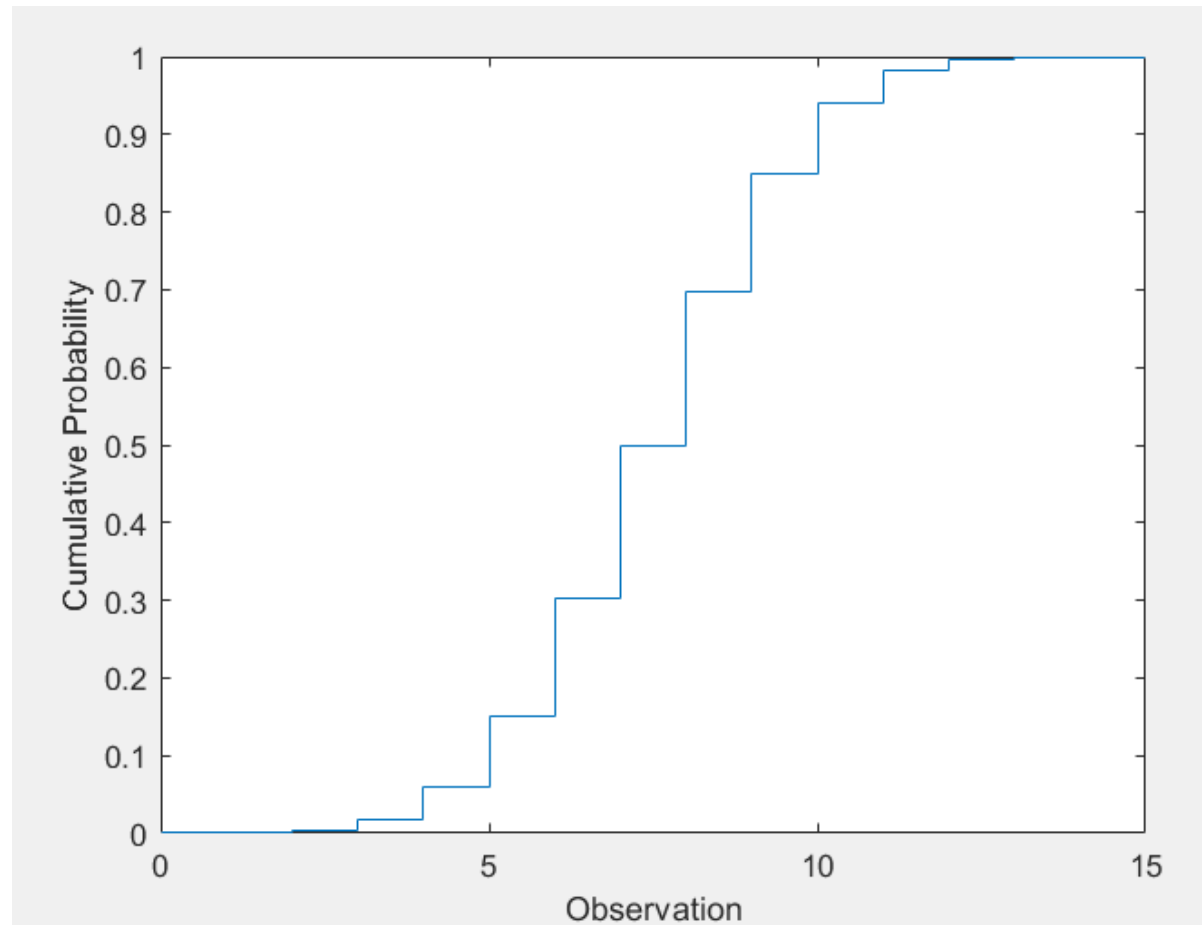
Plotting of CDF:

```
>> figure
```

```
stairs(x,y)
```

```
xlabel('Observation')
```

```
ylabel('Cumulative Probability')
```



Poisson Distribution and its PDF

- The Poisson distribution is a one-parameter family of curves that models the number of times a random event occurs.

Compute the PDF of the Poisson distribution with parameter $\lambda = 5$.

```
>> x = 0:15
```

```
>> y = poisspdf(x,5)
```

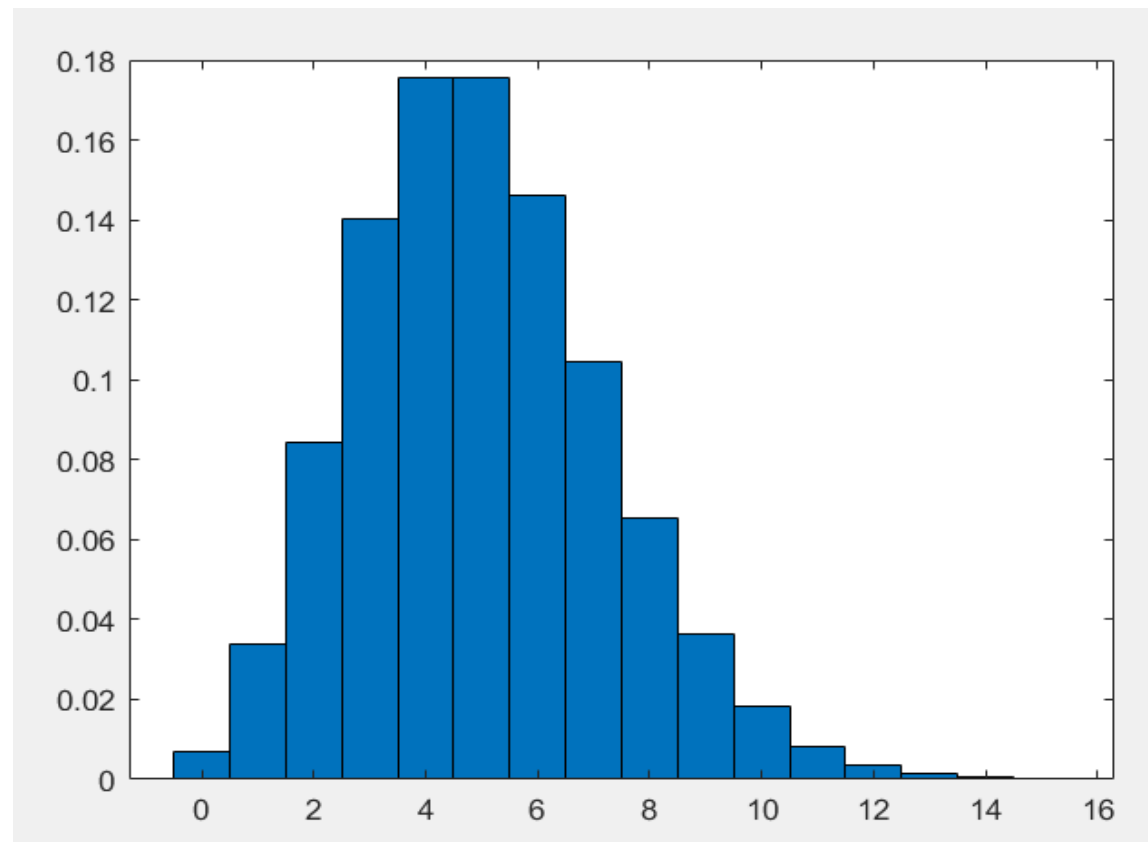
Plotting of PDF with bars of width 1:

```
>> figure
```

```
bar(x,y,1)
```

```
xlabel('Observation')
```

```
ylabel('Probability')
```



Poisson Distribution CDF

```
>> x = 0:15
```

```
>> y = poisspdf(x,5)
```

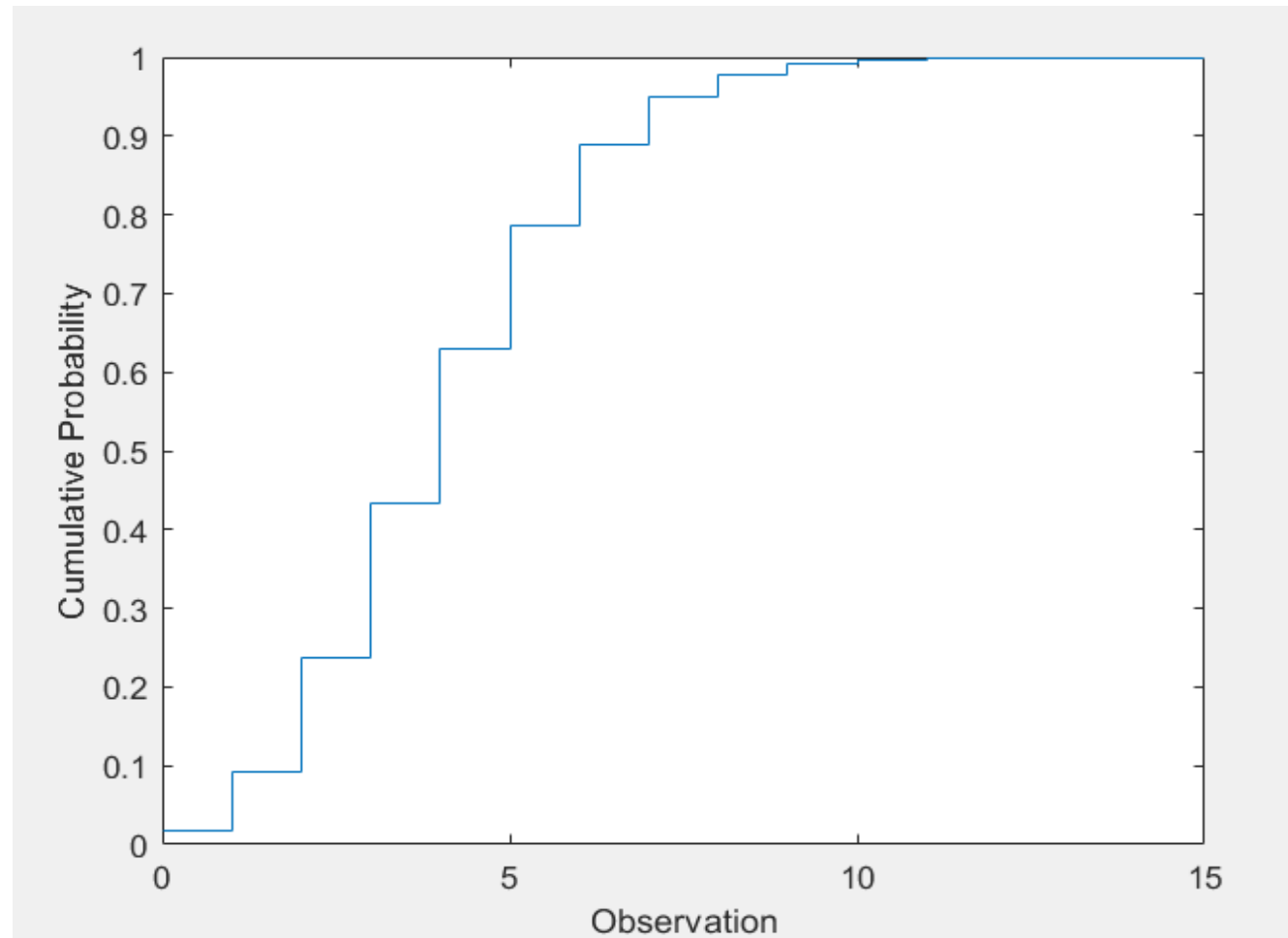
Plotting of CDF:

```
>> figure
```

```
stairs(x,y)
```

```
xlabel('Observation')
```

```
ylabel('Cumulative Probability')
```



Normal Distribution and its PDF

- The normal distribution, sometimes called the Gaussian distribution, is a two-parameter family of curves.

Compute the pdf of a standard normal distribution, with parameters

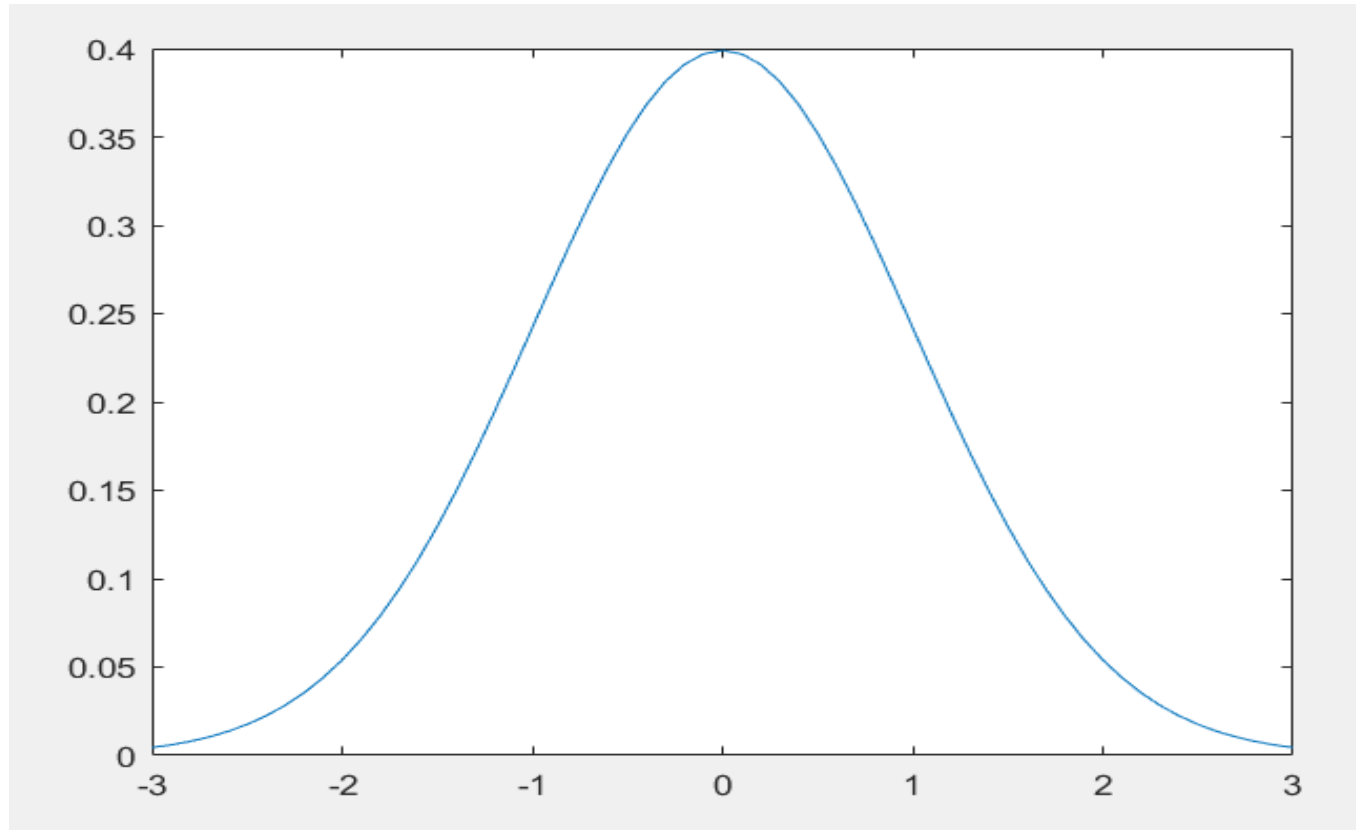
$\mu = 0$; $\sigma = 1$.

```
>> x = [-3:.1:3];
```

```
>> y = normpdf(x,0,1);
```

Plotting of PDF

```
>> plot(x,y)
```



Normal Distribution CDF

```
>> pd = makedist('Normal')
```

```
pd = NormalDistribution
```

```
Normal distribution
```

```
mu = 0
```

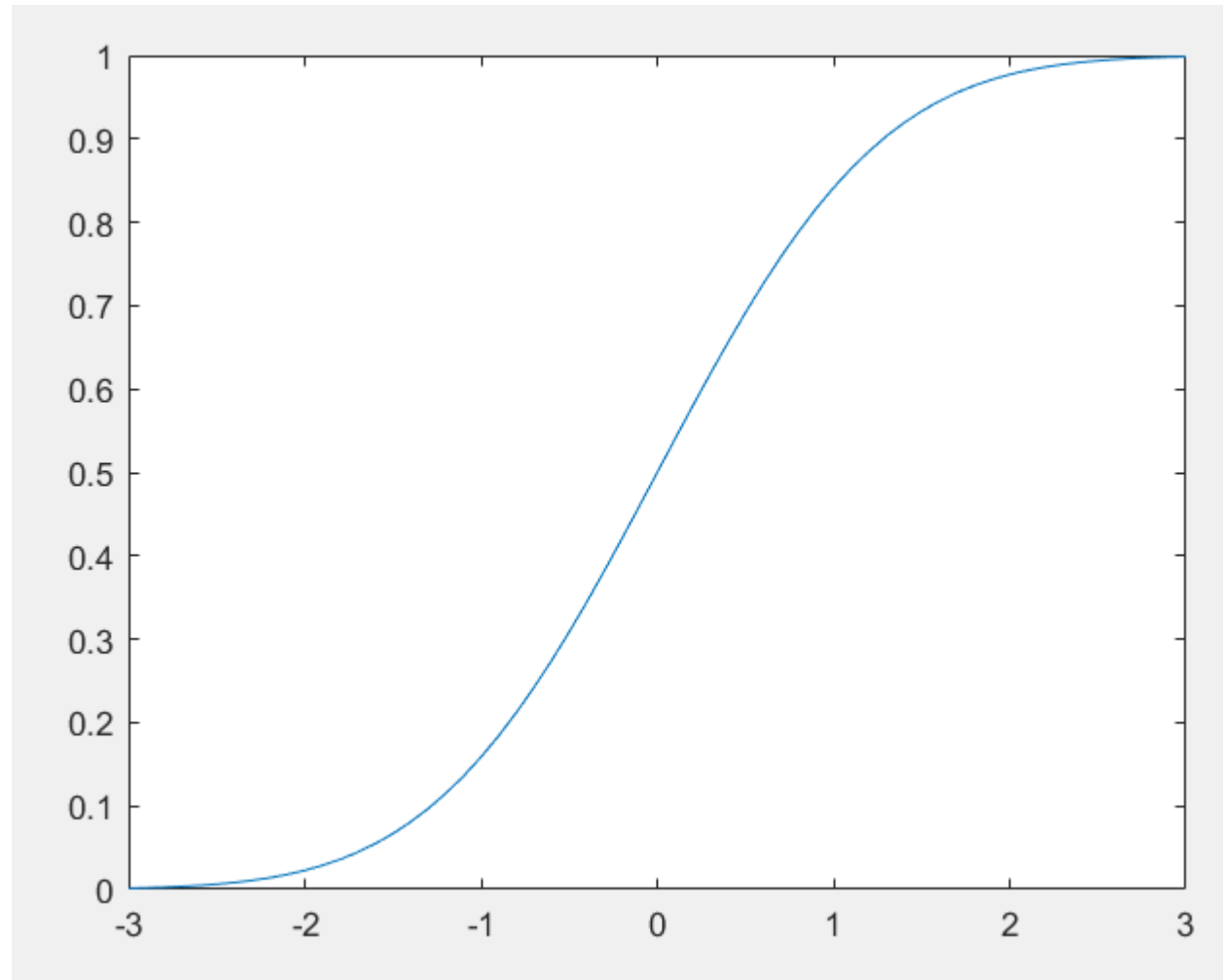
```
sigma = 1
```

```
>> x = -3:1:3;
```

```
>> p = cdf(pd,x)
```

Plotting of CDF

```
>> plot(x,p)
```





THANK YOU
